# GATE

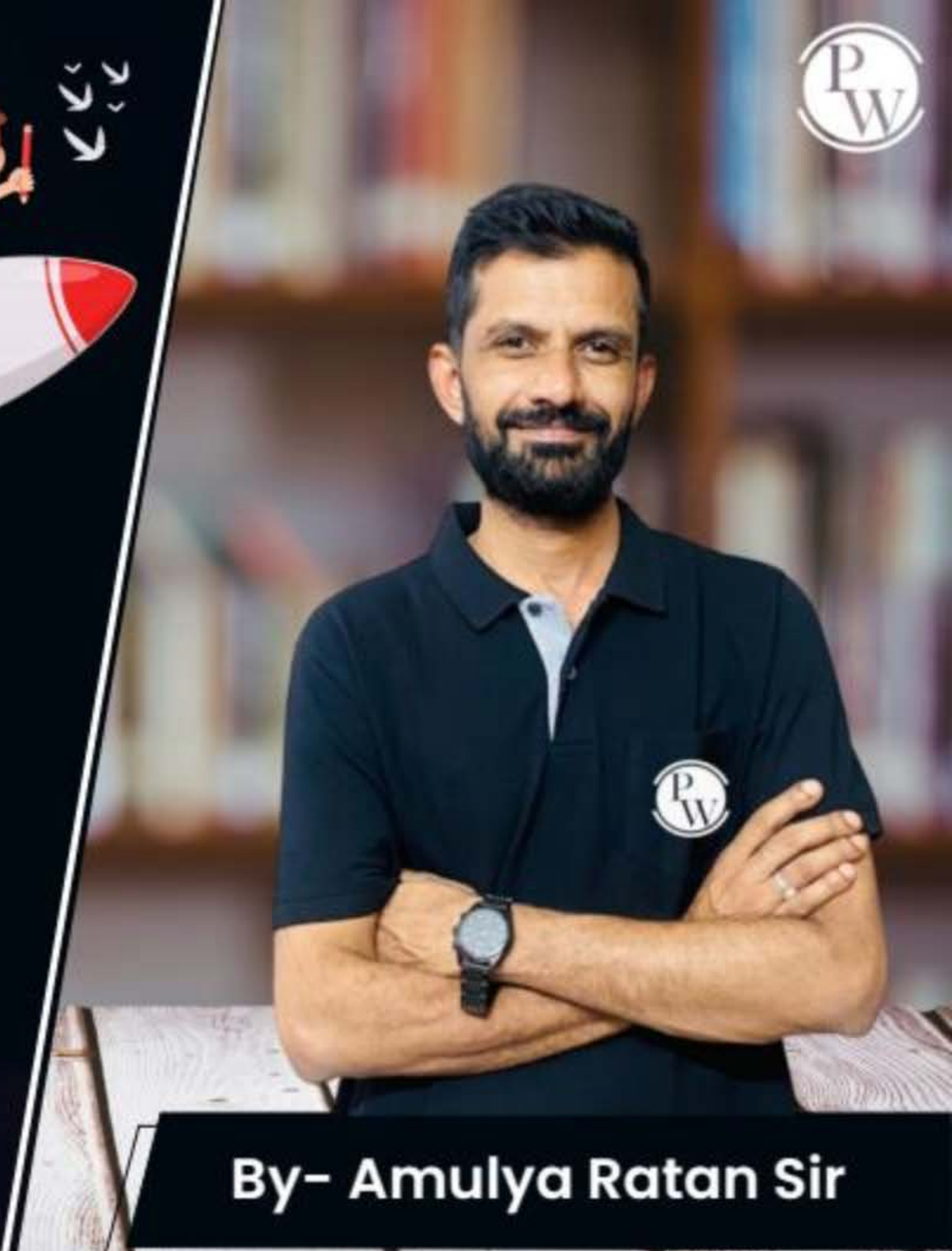
## ALL BRANCHES



## General Aptitude

## QUANTITATIVE APTITUDE

Lecture No.- 16



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# Recap of Previous Lecture



**Topic** Mensuration 2D & 3D

# Topics to be Covered



**Topic-1** Interest

[MCQ]

#Q. The length, breadth & height of a cuboid are in the ratio of 4 : 3 : 2 and its volume is 3000 $m^3$. Find its total surface area.

Assignment

$l = 4x \Rightarrow 4 \times 5 = 20$

$b = 3x \rightarrow 3 \times 5 = 15$

$h = 2x \rightarrow 2 \times 5 = 10$

Volume $= l \times b \times h$

$\Rightarrow 3000 = 4x \times 3x \times 2x$

$\Rightarrow 3000 = 24x^3$

$\therefore x^3 = \frac{3000}{24} = 125$

$\therefore x = \sqrt[3]{125} = 5$

$TSA = 2(lb + bh + hl)$

$= 2(300 + 150 + 200)$

$= 2 \times 650$

$= 1300\ m^2$

A 1300 $m^2$

B 1500 $m^2$

C 1333 $m^2$

D 2700 $m^2$

[MCQ]

#Q. The dimensions of a room are 20 m x 15 m x 10 m. What is the cost of painting its four walls at the rate of ₹ 15 per 100 sq m?

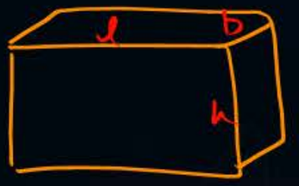
Assignment

$l$ $b$ $h$

$LSA = 2h(l+b)$

$= 2 \times 10 \times 35$

$= 700\ m^2$




$2lh + 2bh$

Cost of painting $= 7 \times 15 = ₹105$

A ₹ 70

B ₹ 105

C ₹ 225

D None of these

[MCQ]

#Q. In two concentric circles, a chord length 80 cm of larger circle becomes a tangent to the smaller circle whose radius is 9 cm. The radius of the larger circle will be?

A 13 cm

B 41 cm

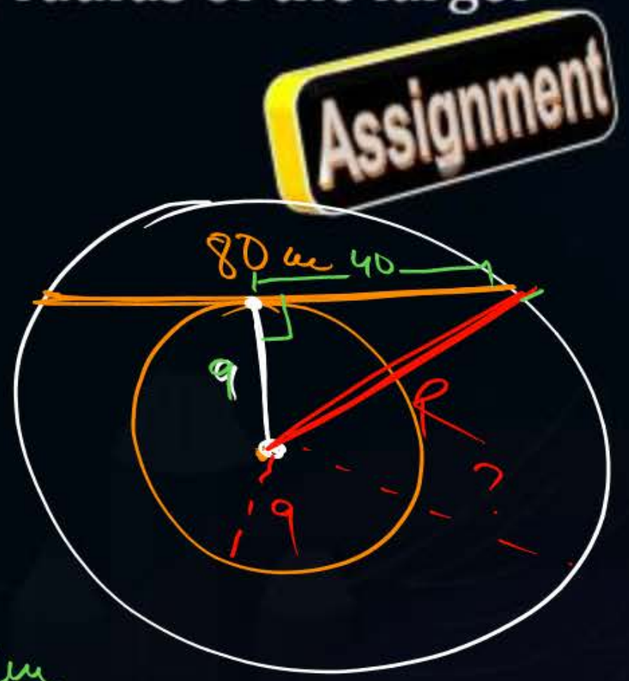
C 52 cm

D 75 cm

$R^2 = 9^2 + 40^2$

$R^2 = 81 + 1600$

$\therefore R = \sqrt{1681} = 41$ cm



[MCQ]

$(a+b)^2 = a^2+b^2+2ab$

$(a+b)^2 - 2ab = a^2+b^2$

#Q. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq cm. The perimeter of the rectangle is?

$$P = 2(l+b)$$

Assignment

$\Rightarrow 2 \times 9 = 18$ cm

A 9 cm

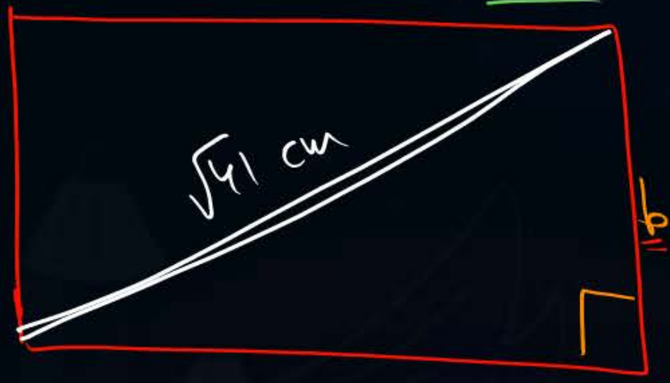
B 18 cm

C 20 cm

D 41 cm

Area of rect. $= l \times b$

$\Rightarrow 20 = lb$




$l^2+b^2 = (\sqrt{41})^2$

$\Rightarrow l^2+b^2 = 41$

$(l+b)^2 - 2lb = 41$

$\Rightarrow (l+b)^2 = 81$

$l+b = \sqrt{81} = 9$

# INTEREST

What?

Why?

$A = P + I$

S.I $\longrightarrow$ 120

C.I $\Rightarrow$ 121

# Simple Interest

$$A = P + I$$

$$I = \frac{TRP}{100}$$

$\pm P$ $\pm P$

$$A = P\left(1 + \frac{TR}{100}\right)$$

# Compound Interest

$T = 3$ years

$A = P\left(1+\frac{R}{100}\right)^n$

Annually $\Rightarrow n = 3$

Half Yearly $\Rightarrow n = 6$

Quarterly $\Rightarrow n = 12$

$I = P\left[\left(1+\frac{R}{100}\right)^n - 1\right]$

$-P$ $\qquad$ $-P$

**Note:**

- S.I OR C.I ?
- ~~S.I.~~ C.I"

# Further formula:

$$P\left(\frac{r}{100}\right)^2$$

Difference between C.I & S.I for two periods

C.I − S.I

$$P\left(1+\frac{r}{100}\right)^2 - P\left(1+\frac{2r}{100}\right)$$

$$\Rightarrow P\left[1+\left(\frac{r}{100}\right)^2+\frac{2r}{100}-1-\frac{2r}{100}\right]$$

[MCQ]

#Q. An investment triples itself in 10 years. In how many years it would be nine times itself?

$A = 3P \Rightarrow 10$ yrs

$I = 2P \Rightarrow 10$ yrs

(×4) (×4)

$I = 8P \Rightarrow 40$ yrs

$A = 9P$

$A = P\left(1 + \frac{TR}{100}\right)$

$3P = P\left(1 + \frac{10R}{100}\right)$

$2 = \frac{R}{10}$

$\therefore R = 20\%$

$9P = P\left(1 + \frac{T \times 20}{100 \times 5}\right)$

$\Rightarrow 8 = \frac{T}{5}$

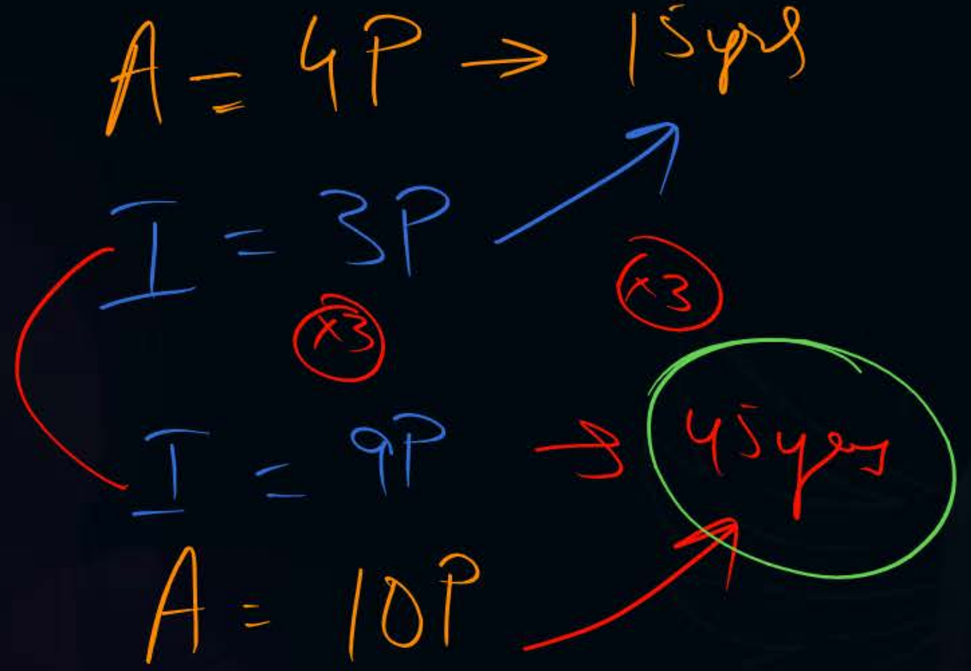
$\therefore$ Time $= 8 \times 5$

$= 40$ yrs

**[MCQ]**

four times

#Q. An investment quadruple itself in 15years. Then in how many years it would be ten times itself?



[MCQ]

P

#Q. An investment doubles itself in 6years on compound interest. Then in how many years it would be sixteen times itself?

$A = P\left(1+\frac{R}{100}\right)^n$

$2P = P\left(1+\frac{R}{100}\right)^{6}$

$()^4$

$16$

$6 \times 4$

$2P \Rightarrow 6\text{ yrs}$

$()^4$

$\times 4$

$16P \Rightarrow 24\text{ years}$

[MCQ]

S.I

#Q. An investment triples itself in 11years. Then in how many years it would be twenty seven times itself?

$A = 3P \rightarrow 11$ yrs

$I = 2P$

$\times 13$

$I = 26P$

$A = 27P \rightarrow$ 143 years

C.I

$3P \rightarrow 11$ yrs

$\times 3$

$\times 3$

$27P \rightarrow 33$ yrs

[MCQ]

#Q. A sum of ₹12000 deposited at compound interest becomes double after 5 years. After 20 years, it will become?

Assignment

- A sum of ₹12000 deposited at compound interest becomes double after 5 years. After 20 years, it will become?

- Sol: The rate of interest at which ₹12000 doubles after 5 years is given by
- $R = 100(x^{\frac{1}{5}} - 1)$
- $= 100(2^{\frac{1}{5}} - 1)$
- $=100 \times (1.1486 - 1)$
- $= 100 \times 0.1486 = 14.86\%$

Therefore, after 20 years it becomes,

$A = ₹12000(1 + 14.86/100)^{20} = ₹12000\ (1.1486)^{20}$

$= ₹12000 \times 15.97 \quad = ₹\ 191671.474$

[MCQ]

#Q. What is the compound interest on ₹2500 for two years at the rate of interest 4% per annum?

Assignment

A ₹ 180

B ₹ 210

C ₹ 204

D ₹ 220

[MCQ]

#Q. A sum of money was invested at simple interest at a certain rate for 3 years. Had it been invested at a 4% higher rate, it would have fetched ₹480 more. Find the principal.

Assignment

A ₹ 3000

B ₹ 4000

C ₹ 5000

D ₹ 6000

[MCQ]

#Q. In what time a sum of money will double itself at a rate of S.I. of 6% per annum?

Assignment

A. 6 ⅓ years

B. 16 ⅔ years

C. 33 ⅓ years

D. 21 years

[MCQ]

#Q. In how many years will a sum of ₹800 at 10% per annum compounded semi-annually become ₹926.10?

Assignment

A. 1 ⅓ years

B. 1 ½ years

C. 2 ⅓ years

D. 2 ½ years

[MCQ]

#Q. The difference between simple interest and compound interest on a sum for 2 years at 8% when the interest is compounded annually is ₹16. If the interest were compounded half yearly the difference in two interests would be nearly

Assignment

A ₹ 21.25

B ₹ 24.64

C ₹ 27.85

D ₹ 29.94

[MCQ]

#Q. Raju invested Rs.5488 for 3 years at C.I. and received an amount of Rs.6750 on maturity. What is rate percent?

Assignment

A $7\frac{1}{7}\%$

B $9\frac{2}{3}\%$

C $3\frac{1}{3}\%$

D $6\frac{1}{3}\%$



# 2 mins Summary

**Topic** Interest

S.I

C.I

# THANK - YOU